COURSE TITLE : PRINTING MACHINERY MAINTENANCE
COURSE CODE : 5102
COURSE CATEGORY : A
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 55
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Maintenance Management	18
2	Power transmission	13
3	Mechanical and Electrical Elements	14
4	Lubrication and Reconditioning.	10
TOTAL		55

Course Outcomes:

G.O	ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENTS WILL BE ABLE:
1	To know different maintenance programs and safety precautions
2	To understand different drives used for power transmission.
3	To know different electrical and mechanical elements used in printing machine
4	To understand machine lubrication, reconditioning and maintenance.

MODULE I MAINTENANCE MANAGEMENT

1.1.0. To know different maintenance programs and safety precautions.

1.1.1. To define what is maintenance.
1.1.2. To explain different types of maintenance.
1.1.3. To know safety precautions and housekeeping.

MODULE II POWER TRANSMISSION

2.1.0 To understand different drives used for power transmission.

2.1.1 To illustrate different chain drives, Belt drives and gear drives
2.1.2 To list different power transmission systems
2.1.3 To define maintenance and lubrication of drive systems.

MODULE III MECHANICAL AND ELECTRICAL ELEMENTS.

3.1.0 To know different electrical and mechanical elements used in printing machine

3.1.1 To identify mechanical and electrical elements of printing machine
3.1.2 To list different bearings, springs and cam and follower
3.1.3 To illustrate different bearings, springs and cam and follower.

MODULE IV LUBRICATION AND RECONDITIONING.

4.1.0 To understand machine lubrication, reconditioning and maintenance.

4.1.1 To define lubrication, machine erection and reconditioning.

4.1.2 To select different lubricants for printing machine.

4.1.3 To list the equipments and tools used for machine erection

4.1.4 To state electrical system, pneumatic system and hydraulic system maintenance.

CONTENT DETAILS

MODULE I

Maintenance Management

Maintenance – Definition , Objectives and functions. Types of Equipment Maintenance – Planned maintenance and unplanned maintenance.
Planned maintenance-Types-Preventive Maintenance, Predictive Maintenance, and Scheduled maintenance - Merits and demerits.
Unplanned maintenance - Breakdown Maintenance or Emergency maintenance - Merits and Demerits.
Contract maintenance - Definition - Merits and Demerits.
Preventive Maintenance – Advantages, Functions - Planning, scheduling - Repair cycles, Dispatching and Controlling. Categories -Inspection, Small repair, Medium repair and Overhauling.
Safety Precautions and House Keeping – safety precautions to be followed in press area - Aids to good housekeeping.

MODULE II

Power Transmission

Chain Drives - Introduction, Types of Chains(Briefly) – Roller Chain, Silent Chain, Ewart Chain and Bead Chain - Merits and Demerits of Chain Drives.
Belt Drives - Introduction, Types of Belts(Briefly) – Flat belt, Rope belt, Tooth Belt, V belt and Timing Belt - Merits and Demerits of Belt drives.
Gear Drives - Introduction, Types of Gears(Briefly) – Spur gear, Helical gear, Bevel gear, Worm gears and Herringbone gear - Merits and Demerits of gear drives.
Maintenance and Lubrication of Drive Systems - Chain Drive, Belt Drive, and Gear Drive.
Direct drive technology - Introduction – Advantages - Application in the printing field.

MODULE III

Mechanical and Electrical Elements

Bearings – Introduction, Merits and Demerits. Types of Bearings (Briefly) - Sliding bearings and Antifriction bearings – Ball bearings, Needle bearings and Roller bearing.

Cams and Follower - Advantages of cam and Follower. Types of Cams and Followers (Briefly) – Disk Cam, Translation Cam, Groove Plate Cam, Cylindrical Cam, Eccentric Cam and Tow Wipe Cam.
Springs - Types of springs (Briefly) – Helical Spring, Conical spring, Volute Spring and Torsion Springs and its application.
Electrical Elements: Introduction to Contactors and its application. Introduction to Limit Switches and its application. Introduction to over Load Relay Switches and its types - Thermal and Magnetic .Introduction to Sensors and Detectors and its application. Introduction to Fuse and its application. Introduction to Electrical Panels.

MODULE IV

Lubrication and Reconditioning.

Lubrication – Introduction - Advantages - Types of Lubricants- Solid, Semisolid and Liquid. Lubrication Schedule-Chart and Paint Marks.
Methods of Lubrication- Gravity feed method-Wick feed, oil feed and Drip Feed , Force feed or Pressure feed method - Oil can, grease gun and grease cup, Splash method - Ring oiler, and Continuous/Multipoint flow method.
Machine Erection and Reconditioning- Introduction, Equipments and Tools used for it, and application.
Electrical Maintenance: AC and DC motors – Introduction - Maintenance Check list for motors, Common problems with Electricity.
Pneumatic System Maintenance: Introduction to pneumatic system functioning – Compressor types - Reciprocating and Rotary compressor - Application in Printing Field . Pump- Definition, application in printing m/c.
Hydraulic System Maintenance - Introduction to Hydraulic System - Application in Printing field.

Reference :

Author	**Title**	**Publishers**
GATF	Lithographers manual	GATF , USA
Herschel L Apfelberg	Maintaining printing equipment	GATF
Barbera, L Albinini	Solving web offset press problem	GATF
HP Garg	Industrial maintenance	S Chand and company ltd